

Analysis of Covid-19 chest x-rays

Data Science Project Report

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Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background and Motivation

COVID-19 is a respiratory disease caused by the SARS-CoV-2 virus. During the pandemic, rapid and reliable diagnosis was an important challenge. Chest X-ray analysis has been explored as a possible tool to support diagnosis because abnormalities in lung structures can appear in infected patients.

1.2 Problem Statement

This project investigates the automated classification of chest X-ray images into four diagnostic categories: COVID-19, Normal, Lung Opacity, and Viral Pneumonia. Chest X-ray imaging is widely used as a fast and accessible diagnostic tool for respiratory diseases, making automated image classification a valuable application of deep learning. The objective of this project is to develop and evaluate convolutional neural network models for multi-class classification while assessing their robustness and interpretability. The dataset consists of publicly available chest X-ray images collected from several repositories and includes corresponding lung segmentation masks. Details of the dataset and its characteristics are presented in Chapter ??.

The project focuses on image classification and does not aim to provide a medical diagnosis or replace healthcare professionals. Furthermore, the objective is not to localize abnormalities within the images.

Chapter 2

Dataset

2.1 Overview

The dataset consists of chest X-ray (CXR) images belonging to four diagnostic categories: *COVID-19*, *Normal*, *Lung Opacity*, and *Viral Pneumonia*. All images are provided in PNG format with a resolution of 299×299 pixels. In addition, lung segmentation masks are available for every image. The dataset was compiled from several publicly available repositories and published studies, with detailed source information provided in the accompanying metadata.

According to the dataset documentation, the four classes were collected from different source repositories rather than a single unified acquisition process. Table ?? summarizes the origin of each class.

Table 2.1: Sources of the chest X-ray images for each class.

Class	Source(s)
COVID-19	PadChest; German Medical School; SIRM/GitHub/Kaggle/Twitter; another GitHub source
Normal	RSNA Pneumonia Detection Challenge (8,851) + Kaggle Chest X-ray Pneumonia Dataset (1,341)
Lung Opacity	RSNA Pneumonia Detection Challenge only
Viral Pneumonia	Kaggle Chest X-ray Pneumonia Dataset only

The COVID-19 class was created by aggregating confirmed COVID-19 chest X-ray images from multiple public repositories to increase the number of available samples. Normal images were obtained primarily from the RSNA Pneumonia Detection Challenge dataset, with additional normal cases collected from the Kaggle Chest X-ray Pneumonia dataset. Lung Opacity images originate exclusively from the RSNA dataset and represent non-COVID pulmonary opacities. Viral Pneumonia images were obtained exclusively from the Kaggle Chest X-ray Pneumonia dataset.

Although combining multiple repositories increases the diversity of the dataset, it also introduces potential biases resulting from differences in image acquisition protocols, imaging equipment, preprocessing pipelines, compression methods, image quality, and patient populations. In particular, all Lung Opacity images originate from the RSNA

dataset, whereas all Viral Pneumonia images are drawn exclusively from the Kaggle dataset. Consequently, a classifier may learn source-specific characteristics instead of disease-related radiographic patterns.

This issue is commonly referred to as *shortcut learning*, where a model exploits acquisition- or dataset-specific artifacts rather than clinically relevant pathological features. To ensure that the trained model focuses on meaningful anatomical regions, explainable artificial intelligence (XAI) techniques such as Grad-CAM should be employed to verify that predictions are based on disease-related radiographic features instead of scanner- or dataset-specific artifacts.

2.2 Dataset Composition and Class Balance

The dataset consists of 21,165 chest X-ray (CXR) images distributed across four diagnostic categories: *COVID-19*, *Lung Opacity*, *Normal*, and *Viral Pneumonia*. Each chest X-ray image has a corresponding lung segmentation mask, resulting in a one-to-one mapping between images and masks.

Table ?? summarizes the distribution of images across the four diagnostic classes. The dataset exhibits a noticeable class imbalance. The *Normal* class is the largest, accounting for 48.15% of all images, followed by *Lung Opacity* (28.41%), *COVID-19* (17.08%), and *Viral Pneumonia* (6.35%).

Table 2.2: Distribution of chest X-ray images across diagnostic classes.

Class	Images	Percentage
COVID-19	3,616	17.08%
Lung Opacity	6,012	28.41%
Normal	10,192	48.15%
Viral Pneumonia	1,345	6.35%
Total	21,165	100.00%

Figure ?? illustrates the class distribution within the dataset.

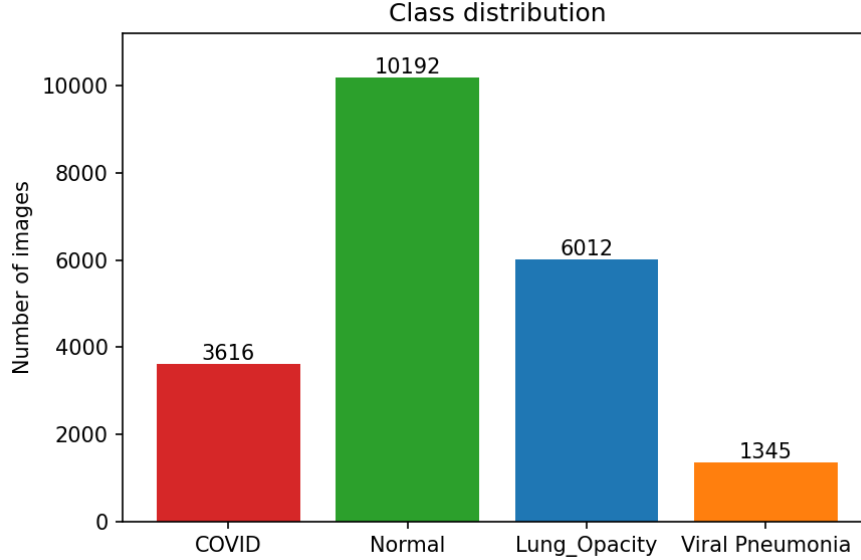


Figure 2.1: Distribution of chest X-ray images across the four diagnostic classes.

To determine whether the observed class frequencies differed significantly from an equal class distribution, a chi-square goodness-of-fit test was performed. The test indicated a highly significant deviation from uniformity ($\chi^2 = 8110.78$, $p < 0.001$). Furthermore, the effect size was large (Cohen’s $w = 0.619$), confirming the presence of substantial class imbalance.

The observed imbalance has important implications for model development and evaluation. During dataset partitioning, stratified sampling should be employed to preserve the original class proportions across the training, validation, and test sets. Furthermore, overall classification accuracy alone may provide a misleading assessment of model performance. Instead, evaluation should include class-sensitive metrics such as per-class precision, recall, macro-averaged F1-score, balanced accuracy, and the confusion matrix to ensure that performance is assessed fairly across all diagnostic categories.

2.3 Structural Integrity, Dimensions, and Image Encoding

To verify the integrity of the dataset, all chest X-ray images and lung segmentation masks were examined prior to model development. Every one of the 21,165 chest X-ray images had a corresponding segmentation mask with an identical filename, indicating a complete one-to-one correspondence between images and masks. Furthermore, no unmatched image-mask pairs or unreadable PNG files were identified.

The average X-ray file size varied slightly across the four diagnostic classes. The mean file sizes were 35.00 KB for COVID-19, 34.40 KB for Lung Opacity, 36.39 KB for Normal, and 39.36 KB for Viral Pneumonia. The relatively larger mean file size observed for the Viral Pneumonia class is explained by the presence of RGB images within this class, which are substantially larger than their grayscale counterparts.

Table ?? summarizes the structural properties of the dataset.

Table 2.3: Structural properties of the chest X-ray images and segmentation masks.

Property	X-rays	Masks
Number of files	21,165	21,165
Dimensions	299×299 px	256×256 px
L mode	21,025	0
RGB mode	140	21,165

Although every image has a corresponding segmentation mask, the image and mask dimensions differ. All chest X-ray images have a resolution of 299×299 pixels, whereas all segmentation masks have a resolution of 256×256 pixels. Consequently, any pixel-wise operation involving both the X-ray image and its corresponding mask requires the mask to be resized or otherwise aligned with the image dimensions.

2.3.1 RGB Images in the Viral Pneumonia Class

A detailed inspection of image encoding revealed that all 140 RGB images are contained exclusively within the Viral Pneumonia class. These RGB images represent 10.41% of the Viral Pneumonia samples (140 of 1,345 images), while every image in the remaining three diagnostic classes is stored as a single-channel grayscale (L-mode) image. Consequently, image encoding is associated with the diagnostic class in the raw dataset.

Table ?? summarizes the characteristics of the Viral Pneumonia images according to image encoding.

Table 2.4: Comparison of grayscale (L) and RGB images within the Viral Pneumonia class.

Mode	n	Mean (KB)	SD	Median	Range (KB)
L	1,205	37.45	2.54	37.47	26.37–47.27
RGB	140	55.82	5.86	56.28	39.83–71.42

Figure ?? compares the mean PNG file sizes of grayscale and RGB images within the Viral Pneumonia class.

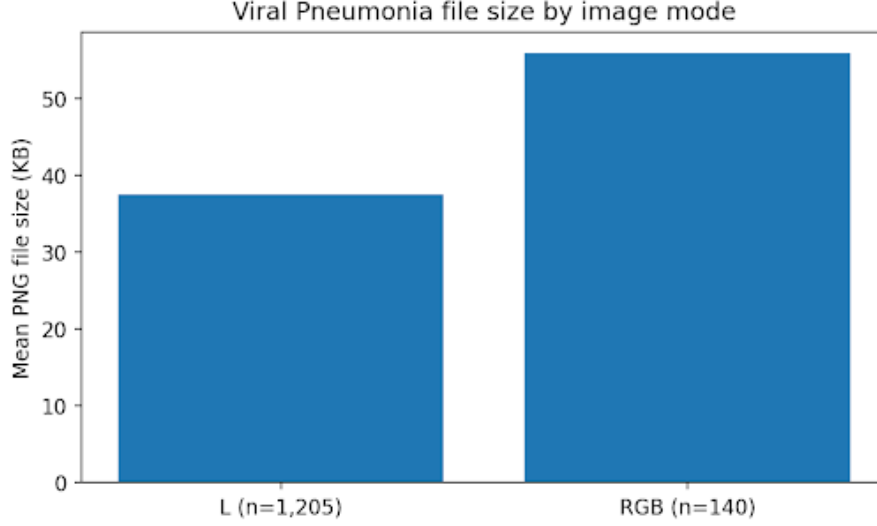


Figure 2.2: Mean PNG file size of Viral Pneumonia images stratified by image encoding.

As shown in Figure ??, RGB images are substantially larger than grayscale (L-mode) images within the same diagnostic category. This finding indicates that the elevated average file size observed for the Viral Pneumonia class is primarily attributable to differences in image encoding rather than disease-related image characteristics.

Since image encoding represents a technical characteristic rather than a clinically meaningful feature, all chest X-ray images should be converted to a common grayscale representation during preprocessing. Standardizing the image format eliminates a potential source of dataset bias and prevents the model from exploiting encoding-related differences as shortcut features.

2.4 Exact Duplicate Analysis

To identify redundant images, perceptual hashing (pHash) was applied using a Hamming-distance threshold of zero. This initial screening identified 287 candidate duplicate image pairs. A subsequent pixel-wise comparison confirmed that 91 of these pairs were exact pixel duplicates, whereas the remaining 196 pairs shared an identical perceptual hash but differed at the pixel level. All confirmed exact duplicates were found within the same diagnostic class, and no duplicate images were shared across different class labels.

Duplicate images were subsequently grouped into connected duplicate sets, and a single representative image was retained from each group. This process identified a total of 59 redundant files, corresponding to only 0.28% of the entire dataset. The distribution of redundant files across diagnostic classes is presented in Table ??.

Table 2.5: Redundant exact duplicate files identified after grouping connected duplicate sets.

Class	Redundant Files	Percentage of Class
COVID-19	51	1.41%
Lung Opacity	0	0.00%
Normal	1	0.01%
Viral Pneumonia	7	0.52%
Total	59	0.28%

Figure ?? illustrates the number of confirmed redundant images identified within each diagnostic class.



Figure 2.3: Distribution of confirmed redundant exact duplicate files by diagnostic class after grouping duplicate sets.

The identified duplicate images represent unnecessary redundancy within the dataset and, if retained, could introduce data leakage by allowing identical images to appear in both the training and evaluation sets. To prevent this issue, the 59 redundant files were removed prior to performing the training/validation/test split, while retaining one representative image from each exact-duplicate group. This preprocessing step helps ensure an unbiased evaluation of model performance and prevents overly optimistic performance estimates resulting from duplicate observations.

2.5 Whole-Image Brightness and Contrast

To characterize global image intensity, two descriptive features were calculated for each chest X-ray. *Brightness* was defined as the mean grayscale pixel intensity of the image, while *contrast* was represented by the standard deviation of grayscale pixel intensities within the image. To ensure consistency, all chest X-ray images were converted to grayscale prior to these calculations.

Table ?? summarizes the descriptive statistics for brightness and contrast across the four diagnostic classes. COVID-19 images exhibited the highest mean brightness (139.52), whereas Normal images showed the highest average global contrast (61.56). Lung Opacity and Viral Pneumonia displayed similar central brightness values, although their contrast differed slightly.

Table 2.6: Descriptive statistics of whole-image brightness and contrast by diagnostic class.

Class	Brightness			Contrast		
	Mean	SD	Median	Mean	SD	Median
COVID-19	139.52	25.03	142.59	54.65	14.83	54.05
Lung Opacity	126.02	23.57	120.82	57.40	9.81	56.97
Normal	129.39	22.45	125.51	61.56	9.55	63.53
Viral Pneumonia	125.37	18.98	126.86	58.75	10.78	59.71

Figure ?? presents the distributions of brightness and contrast for each diagnostic class. COVID-19 images tend to exhibit higher overall brightness, whereas the Normal class has the highest global contrast. The boxplots also reveal considerable within-class variability and numerous statistical outliers, particularly for COVID-19. Since these measurements were computed over the entire image, they may also be influenced by background regions outside the lungs. Consequently, evaluating brightness and contrast within the segmented lung regions may provide a more clinically meaningful assessment.

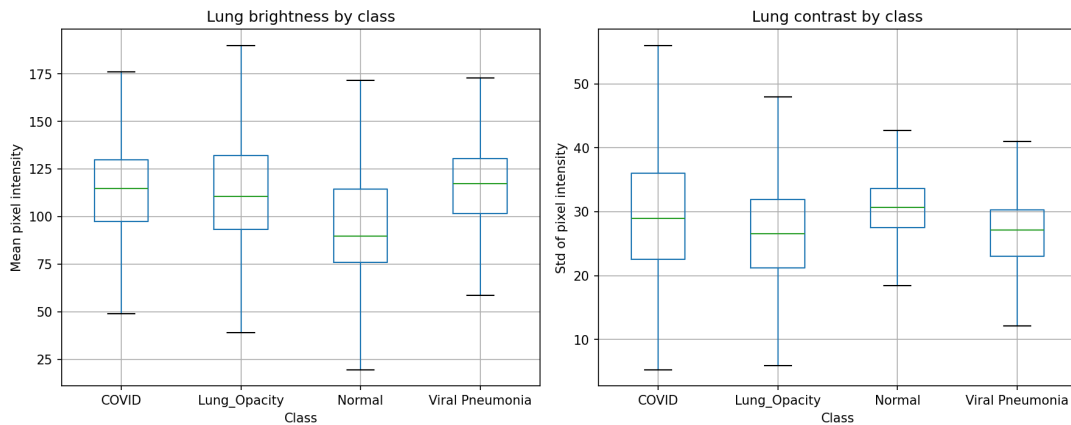


Figure 2.4: Whole-image brightness and contrast distributions by diagnostic class.

Figure ?? illustrates the relationship between whole-image brightness and contrast. Although class-specific tendencies are visible, substantial overlap exists among all four diagnostic classes. COVID-19 images exhibit a broader spread in both brightness and contrast, whereas Viral Pneumonia images are more tightly clustered. Normal images tend toward higher contrast, consistent with the descriptive statistics shown in Table ?. The considerable overlap indicates that global intensity characteristics alone are unlikely to provide sufficient discriminative information for reliable disease classification.

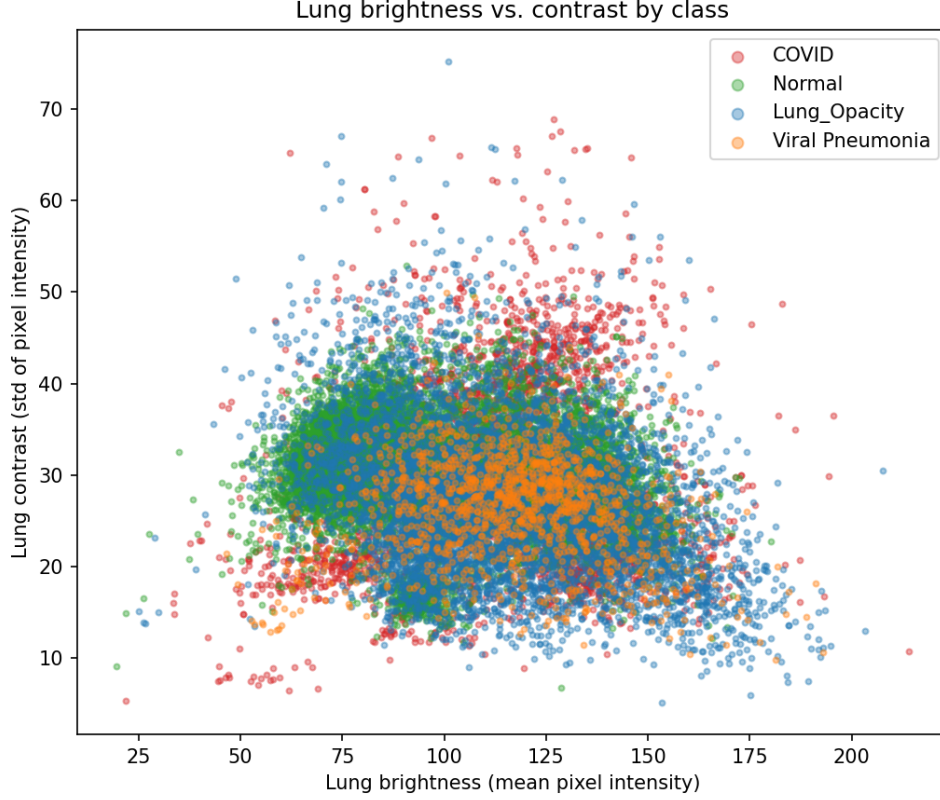


Figure 2.5: Relationship between whole-image brightness and contrast by diagnostic class.

The observed differences in global brightness and contrast are more likely to reflect variations in image acquisition protocols, scanner characteristics, and preprocessing procedures than disease-specific radiographic features. Consequently, these global intensity characteristics may represent technical confounding factors that a machine learning model could exploit as *shortcut features*. Applying intensity normalization during preprocessing and evaluating its impact on model performance can help determine whether the classifier relies on clinically meaningful radiographic patterns rather than global intensity differences.

2.5.1 Statistical Testing

Non-parametric statistical analyses were performed to compare brightness and contrast across the four diagnostic classes. A Kruskal–Wallis test indicated significant differences in whole-image brightness ($H(3) = 987.39$, $p < 0.001$), with a small overall effect size ($\varepsilon^2 = 0.0465$). Significant differences were also observed for whole-image contrast ($H(3) = 1393.78$, $p < 0.001$), with a modest effect size ($\varepsilon^2 = 0.0657$).

To quantify the magnitude of pairwise differences, Cohen’s d was computed for each pair of diagnostic classes. The resulting effect sizes are summarized in Table ??.

Table 2.7: Pairwise Cohen’s d effect sizes for whole-image brightness and contrast.

Comparison	Brightness	Contrast
COVID-19 vs Lung Opacity	0.559	-0.230
COVID-19 vs Normal	0.438	-0.618
COVID-19 vs Viral Pneumonia	0.601	-0.296
Lung Opacity vs Normal	-0.147	-0.431
Lung Opacity vs Viral Pneumonia	0.029	-0.135
Normal vs Viral Pneumonia	0.182	0.290

Post hoc Dunn tests with Bonferroni correction revealed significant brightness differences for every class pair except Lung Opacity versus Viral Pneumonia. Although the comparison between Normal and Viral Pneumonia remained statistically significant (adjusted $p = 0.0012$), the corresponding effect size was small ($d = 0.182$). For whole-image contrast, all six pairwise comparisons remained statistically significant following Bonferroni adjustment. The largest standardized difference was observed between COVID-19 and Normal ($d = -0.618$), indicating that COVID-19 images generally exhibit lower global contrast than Normal images.

Overall, these findings demonstrate that brightness and contrast differ systematically across diagnostic classes. However, the relatively small-to-moderate effect sizes and substantial overlap among class distributions suggest that global intensity characteristics alone are insufficient for robust disease classification. Instead, they should primarily be regarded as potential technical confounding variables that require careful preprocessing and interpretation during model development.

2.6 Summary of Current Evidence

Table ?? summarizes the principal findings obtained during the exploratory data analysis and their implications for the subsequent preprocessing and model development stages.

Table 2.8: Summary of key findings from the exploratory data analysis.

Domain	Main Result	Implication
Class balance	$\chi^2(3) = 8110.78$; Cohen's $w = 0.619$	Substantial class imbalance, with the Normal class dominating the dataset and Viral Pneumonia being underrepresented.
Integrity	0 unreadable files; 0 unmatched image-mask pairs	Complete correspondence between chest X-ray images and segmentation masks.
Image encoding	140 RGB X-rays, all belonging to the Viral Pneumonia class	Class-specific technical artifact; all images should be standardized to grayscale before model training.
Exact duplicates	59 redundant files, including 51 COVID-19 images	Duplicate images should be removed prior to dataset partitioning to prevent data leakage.
Lung-mask coverage	$H(3) = 1493.59$; $\varepsilon^2 = 0.0704$	Modest but statistically significant differences in lung-mask coverage across diagnostic classes.
Whole-image brightness	$H(3) = 987.39$; $\varepsilon^2 = 0.0465$	Small but significant class-associated differences in global image brightness.
Whole-image contrast	$H(3) = 1393.78$; $\varepsilon^2 = 0.0657$	Modest but significant class-associated differences in global image contrast.

Overall, the dataset is structurally complete and internally consistent. However, several characteristics were identified that could influence machine learning performance independently of clinically meaningful pathology. These include substantial class imbalance, class-specific RGB image encoding, concentrated exact duplication within the COVID-19 class, statistically significant differences in lung-mask coverage, and systematic differences in whole-image brightness and contrast across diagnostic categories.

Although these characteristics are not inherently problematic, they represent potential technical confounding factors that could encourage a model to learn dataset-specific shortcuts instead of disease-related radiographic features. Consequently, the preprocessing pipeline has been designed to address these issues through deduplication, image standardization, mask alignment, intensity normalization, and stratified dataset partitioning. The effectiveness of these preprocessing steps will be evaluated during model development to ensure that the resulting classifiers learn clinically meaningful representations and generalize reliably to previously unseen data.

Chapter 3

Preprocessing

The exploratory analysis in Chapter ?? identified several characteristics that could bias a classifier independently of pathology: exact duplicate images, class-specific RGB encoding, mismatched image-mask resolutions, class imbalance, and systematic differences in global brightness and contrast. The preprocessing strategy described here addresses these issues before model training while preserving the original dataset and documenting every decision for reproducibility.

3.1 Design Rationale

Each preprocessing choice follows directly from the exploratory findings. Duplicate removal prevents data leakage between training and evaluation sets. Grayscale conversion removes the class-associated RGB encoding found in 10.41% of Viral Pneumonia images. Mask resizing with nearest-neighbor interpolation preserves lung boundaries when aligning 256×256 masks with 299×299 X-rays. Stratified partitioning maintains the imbalanced class distribution across splits. Per-image min-max normalization reduces global intensity differences that may reflect acquisition protocol rather than pathology. Optional contrast enhancement (CLAHE), lung-region masking, and training-time augmentation are retained as controlled alternatives for later comparison.

Figure ?? summarizes the overall workflow from the raw collection of 21,165 images to the modeling-ready set of 21,106 samples.

Raw dataset (21,165 images) $\rightarrow$ remove 59 exact duplicates $\rightarrow$ cleaned dataset (21,106 images) $\rightarrow$ stratified split (70% / 15% / 15%) $\rightarrow$ per-image pipeline (grayscale, resize, optional CLAHE / lung mask / augmentation, min-max normalization) $\rightarrow$ training, validation, and test partitions
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Figure 3.1: End-to-end preprocessing workflow. CLAHE, lung masking, and augmentation were disabled in the baseline configuration but remain available for ablation studies.

3.2 Deduplication and Processed Dataset

Exact duplicate groups identified during exploration yielded 59 redundant files. These were excluded *before* splitting so that identical observations cannot appear in both training and evaluation subsets.

Table ?? reports the class distribution after removal. The relative proportions remain close to the raw dataset because duplicates represented only 0.28% of all images and were concentrated in the COVID-19 class, where 51 of 3,616 images (1.41%) were redundant.

Table 3.1: Class distribution after removing 59 exact duplicate files.

Class	Images	Percentage
COVID-19	3,565	16.89%
Lung Opacity	6,012	28.48%
Normal	10,191	48.29%
Viral Pneumonia	1,338	6.34%
Total	21,106	100.00%

The 196 image pairs that share a perceptual hash but differ at the pixel level were not removed automatically. They remain flagged for manual or similarity-based review because automatic exclusion could discard legitimately distinct acquisitions.

3.3 Stratified Partitioning

The cleaned dataset was divided using a two-step stratified procedure with a fixed random seed to ensure reproducibility:

1. 70% assigned to training and 30% held out for evaluation.
2. The holdout set divided equally into validation and test subsets, yielding an overall 70% / 15% / 15% partition.

Class proportions were preserved at each step. Table ?? lists the resulting sample counts; Table ?? shows that normalized class frequencies remain aligned across all three splits, with a maximum deviation of 0.03 percentage points for Viral Pneumonia.

Table 3.2: Absolute sample counts per split after deduplication and stratification.

Class	Train	Validation	Test	Total
COVID-19	2,495	535	535	3,565
Lung Opacity	4,208	902	902	6,012
Normal	7,134	1,528	1,529	10,191
Viral Pneumonia	937	201	200	1,338
Total	14,774	3,166	3,166	21,106

Table 3.3: Class proportions within each split.

Class	Train	Validation	Test
COVID-19	16.89%	16.90%	16.90%
Lung Opacity	28.48%	28.49%	28.49%
Normal	48.29%	48.26%	48.29%
Viral Pneumonia	6.34%	6.35%	6.32%

3.4 Per-Image Transformation

Each chest X-ray underwent the following sequence of transformations. The order matters: geometric operations precede contrast adjustment and masking, and normalization is applied last so that intensity scaling reflects the final pixel content presented to the model.

1. **Grayscale conversion.** All images were converted to single-channel grayscale, eliminating the RGB encoding bias described in Chapter ??.
2. **Resize.** Images were scaled to 224×224 pixels using area-based interpolation. Segmentation masks were resized to the same dimensions with nearest-neighbor interpolation to preserve binary lung boundaries.
3. **Optional augmentation.** When enabled, geometric and photometric transforms were applied to the image and mask jointly so that spatial correspondence is maintained (Section ??).
4. **Optional CLAHE.** Contrast Limited Adaptive Histogram Equalization (clip limit 2.0, tile size 8×8) was applied to the image only, targeting locally varying contrast without altering mask geometry.
5. **Optional lung masking.** Pixels outside the binarized lung region were zeroed before normalization, restricting intensity statistics to pulmonary tissue.
6. **Per-image min–max normalization.** Each image was scaled to $[0, 1]$ using its own minimum and maximum pixel value:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min} + \varepsilon}, \quad \varepsilon = 10^{-8}.$$

Independent normalization attenuates scanner- and acquisition-specific global brightness differences identified in the brightness–contrast analysis.

Figure ?? illustrates these stages on a representative COVID-19 chest X-ray. The aligned lung mask shows nearest-neighbor resizing from 299×299 to 224×224 . CLAHE and lung masking are shown for illustration; both were disabled in the baseline configuration.

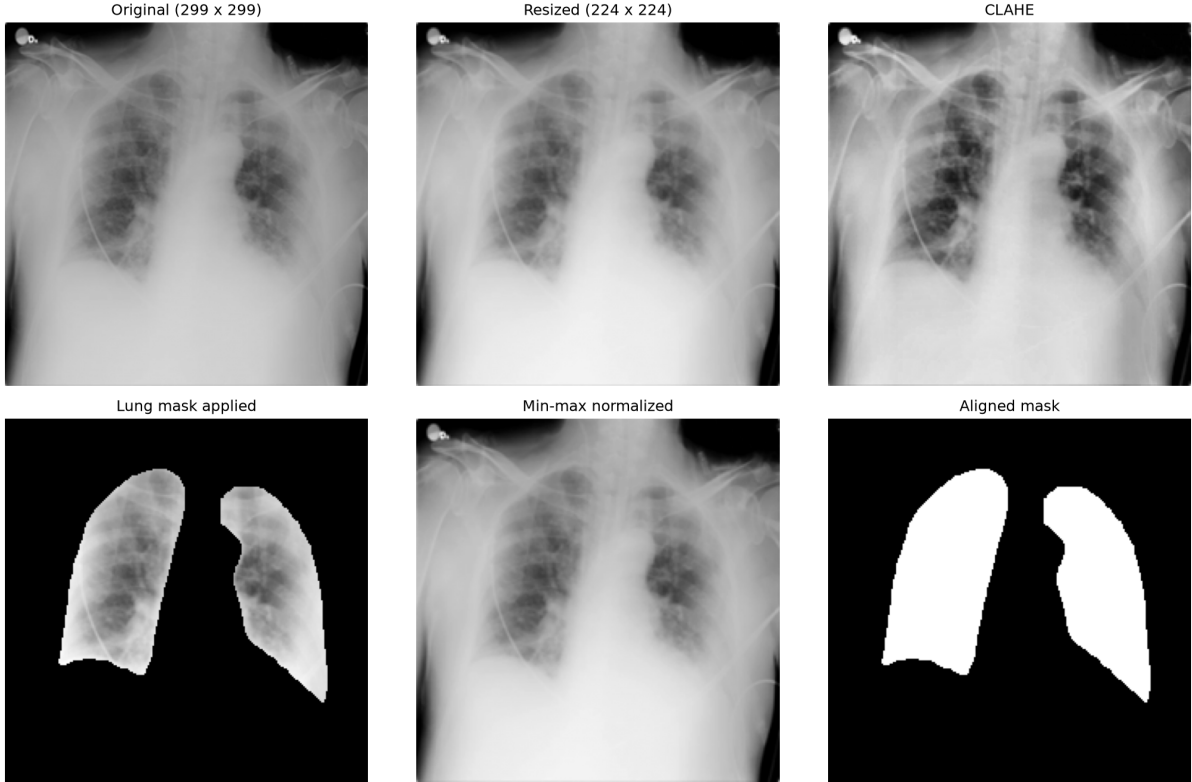


Figure 3.2: Per-image preprocessing stages on a representative COVID-19 chest X-ray: original grayscale image, resize to 224×224 , CLAHE, lung masking, min-max normalization, and aligned segmentation mask.

Table ?? summarizes the baseline configuration. CLAHE, lung masking, and augmentation were disabled so that their effect on downstream classification can be evaluated as controlled ablations.

Table 3.4: Baseline preprocessing configuration.

Setting	Baseline value
Output spatial size	224×224 pixels
Grayscale conversion	Applied to all images
CLAHE	Disabled
Lung mask application	Disabled
Lung mask binarization threshold	127
Training-time augmentation	Disabled
Random seed	42
Train / validation / test ratio	70% / 15% / 15%

3.5 Training-Time Augmentation

Data augmentation was designed to increase effective training-set diversity without breaking the correspondence between chest X-rays and lung segmentation masks. Geometric transforms were applied to both modalities jointly; photometric adjustments affected the

image only. Augmentation was restricted to the training partition so that validation and test performance reflect unmodified acquisitions.

Table ?? lists the operations and their activation probabilities. Each operation is applied stochastically and independently, so a single training sample may receive none, one, or several transforms.

Table 3.5: Augmentation operations and activation probabilities.

Operation	Probability	Effect
Horizontal flip	0.5	Mirror image and mask about the vertical axis
Rotation with zoom compensation	0.5	Up to $\pm 15^\circ$; borders reflected on image
Random zoom	0.5	Scale change up to $\pm 8\%$; center crop or pad
Translation	0.5	Shift up to $\pm 5\%$ of width and height
Brightness / contrast	0.4	Intensity scaling up to $\pm 10\%$; image only

Rotation without zoom compensation was evaluated separately and found to introduce empty border regions in the segmentation mask when reflect padding was used. The zoom-compensated rotation variant scales the image to fill the frame and uses zero padding on the mask, keeping lung labels binary. Figure ?? compares individual geometric transforms on a representative COVID-19 case. Figure ?? shows the combined effect of a single stochastic augmentation pass on the same sample.

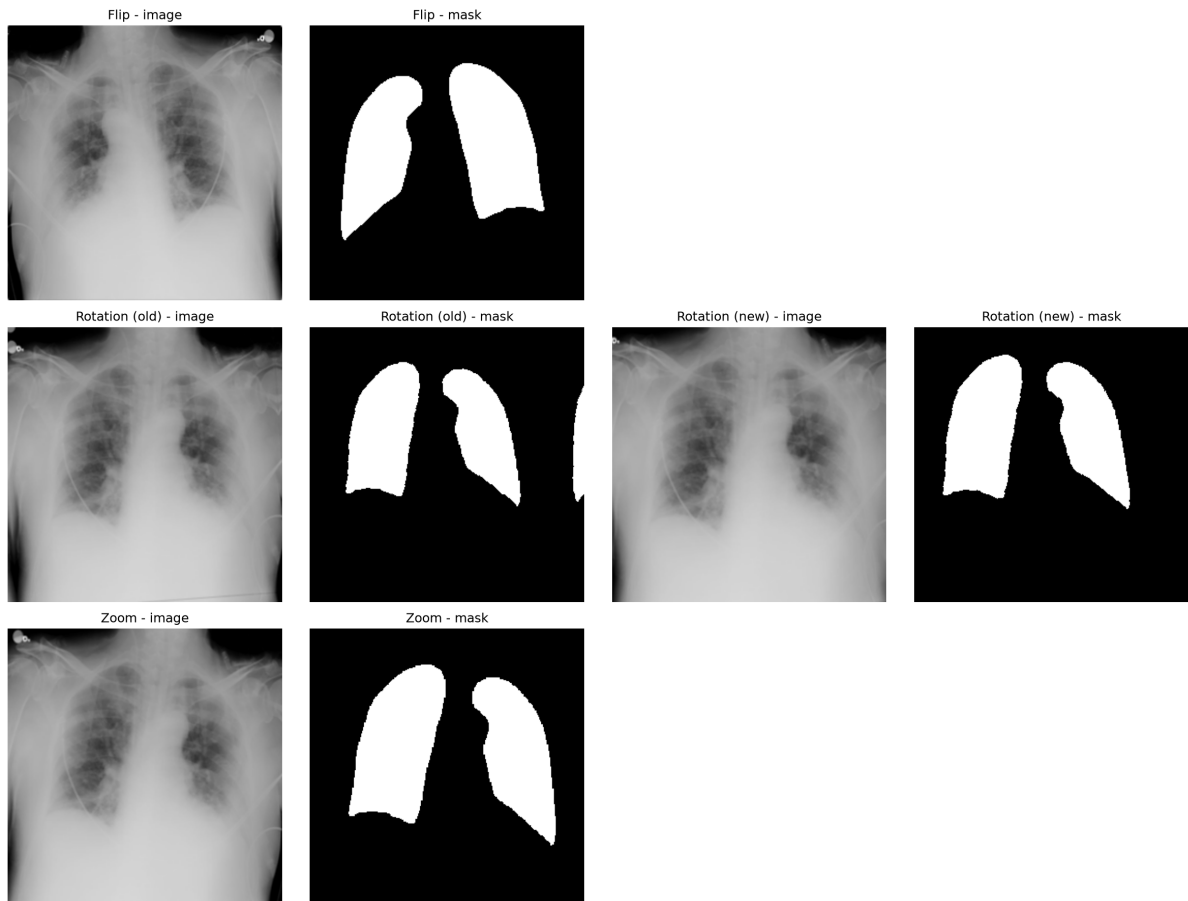


Figure 3.3: Individual augmentation operations on a representative COVID-19 chest X-ray: horizontal flip, rotation without zoom compensation, rotation with zoom compensation, and random zoom. Each row pairs the transformed image with its aligned lung mask.

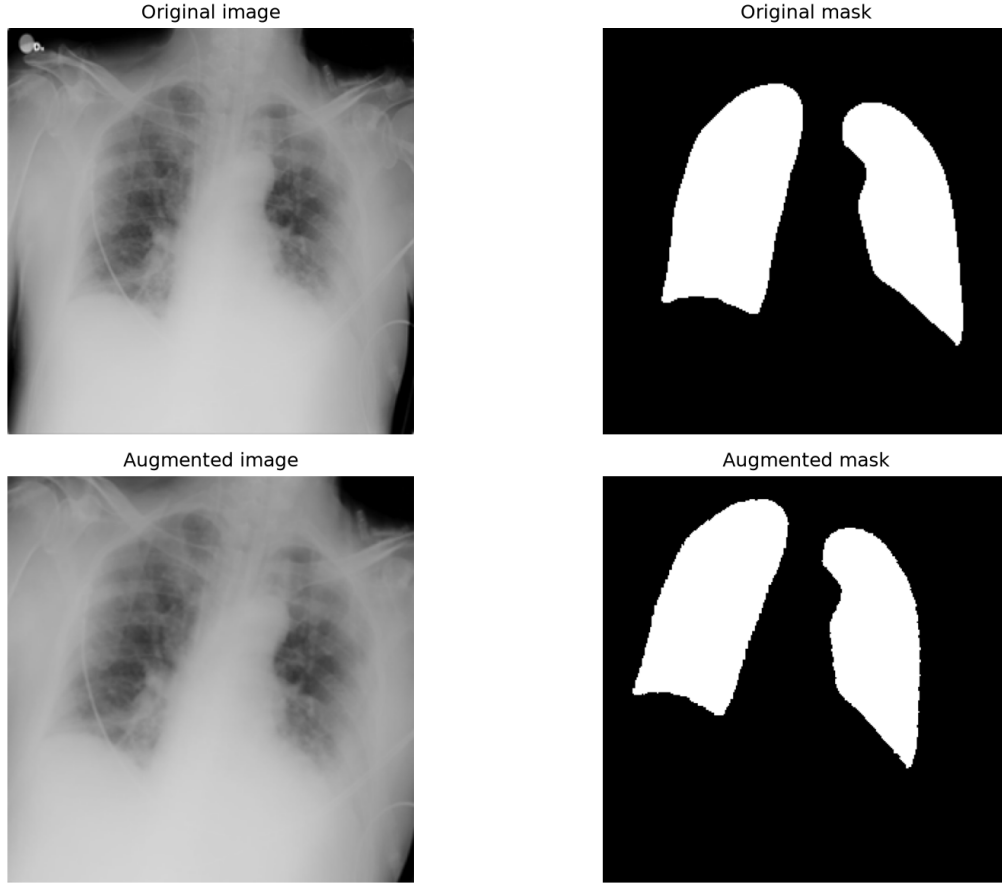


Figure 3.4: Original COVID-19 chest X-ray and lung mask compared with one stochastic augmentation pass (fixed random seed for reproducibility).

3.6 Verification and Reproducibility

The preprocessing workflow was verified at three levels before model training.

First, **transform correctness** was confirmed on synthetic images: normalized outputs lie in $[0, 1]$, repeated processing without augmentation yields identical results, lung masking zeroes non-pulmonary regions, and CLAHE changes pixel values while preserving spatial dimensions. Geometric augmentation preserves image and mask shape across 25 consecutive stochastic draws.

Second, **split integrity** was confirmed: the three partitions are disjoint, together they account for all 21,106 retained samples, and class proportions match the full dataset within 0.05 percentage points.

Third, **end-to-end consistency** was confirmed on the full dataset: all 21,106 images were read successfully with zero failures; processed image counts, split totals, and stacked sample counts all equal 21,106; and a complete rebuild of the baseline configuration produced bit-identical outputs to the saved training, validation, and test partitions ($14,774 + 3,166 + 3,166 = 21,106$).

A total of 78 automated checks covering transforms, augmentation, splitting, and pipeline regression passed. Stochastic steps used random seed 42 throughout.

3.7 Implications for Model Development

This preprocessing stage establishes a fixed, auditable baseline for convolutional network training. Stratified splitting and duplicate removal reduce leakage risk; grayscale standardization and per-image normalization target the technical shortcuts documented in Chapter ???. Because class imbalance persists after splitting—Viral Pneumonia remains at 6.34% of the training set—downstream evaluation must report per-class precision, recall, macro-averaged F1, balanced accuracy, and confusion matrices rather than overall accuracy alone.

Optional CLAHE, lung masking, and augmentation remain available for systematic comparison in subsequent experiments. Their impact on classification performance and on whether the model attends to pulmonary rather than acquisition-related features will be assessed during model development.

Chapter 4

Methodology

Chapter 5

Modelling

5.1 Baseline Models and Initial Explainability Analysis

Two baseline configurations were established before evaluating more advanced approaches such as data augmentation and transfer learning. The objective was not only to obtain reference classification performance, but also to examine whether the convolutional neural network (CNN) relied on non-pulmonary or dataset-specific visual cues.

The first baseline used the complete chest radiograph and represented an unconstrained classification setting. However, additional region-based experiments showed that substantial predictive information was also present outside the segmented lung fields. Therefore, a second baseline was introduced using a lung-centred region of interest (ROI). In this configuration, the lung segmentation masks were used only to localise the thoracic ROI and were not applied directly to the input image. This avoided the artificial black boundaries introduced by hard lung masking while reducing peripheral image content.

The two baselines therefore served complementary purposes:

- **Full-image baseline:** measures the classification performance achievable from the complete chest X-ray.
- **Lung-ROI baseline:** provides a more spatially controlled reference in which peripheral image regions are reduced while the original radiographic appearance is preserved.

Neither baseline used data augmentation or transfer learning.

5.1.1 Experimental Setup

Both baselines used the same deliberately simple CNN trained from scratch. The architecture consisted of three convolutional blocks followed by global average pooling and a four-class softmax classifier:

Simple CNN Architecture

Input: $128 \times 128 \times 1$
Conv2D, 16 filters $\rightarrow$ MaxPooling
Conv2D, 32 filters $\rightarrow$ MaxPooling
Conv2D, 64 filters $\rightarrow$ MaxPooling
Global Average Pooling
Dense layer with 4-class softmax output

The model contained 23,556 trainable parameters. The classification task included four classes: COVID, Lung Opacity, Normal, and Viral Pneumonia. The same duplicated dataset, train/validation/test partitions, random seed, class-weighting strategy, and optimisation settings were maintained across both baseline experiments.

The class distributions were identical for the two baselines and are shown in Table ??.

Table 5.1: Class distribution used for the baseline experiments.

Class	Train	Validation	Test
COVID	16.89%	16.90%	16.90%
Lung Opacity	28.48%	28.49%	28.49%
Normal	48.29%	48.26%	48.29%
Viral Pneumonia	6.34%	6.35%	6.32%

Class weighting was used to reduce the effect of class imbalance. Training was performed for a maximum of 100 epochs with early stopping based on validation loss.

5.1.2 Baseline 1: Full-Image Simple CNN

The first baseline was trained directly on the complete chest radiograph, without augmentation or additional intensity or geometry normalisation.

Its test-set performance is presented in Table ??.

Table 5.2: Test-set performance of the full-image simple CNN baseline.

Class	Precision	Recall	F1-score
COVID	0.766	0.830	0.796
Lung Opacity	0.799	0.766	0.782
Normal	0.850	0.831	0.840
Viral Pneumonia	0.840	0.945	0.889
Macro average	0.813	0.843	0.827

The full-image model achieved a test accuracy of **81.96%** and a macro F1-score of **82.7%**. This represents the strongest raw baseline performance obtained with the simple CNN.

However, the full-image model should be interpreted as a *naive performance baseline*. A separate background-only experiment achieved approximately 73.40% test accuracy,

indicating that non-lung regions contain substantial class-correlated information. Therefore, high full-image classification performance does not necessarily imply that the model relies exclusively on clinically relevant pulmonary features.

5.1.3 Grad-CAM Analysis of the Full-Image Baseline

Grad-CAM was applied to the full-image model to investigate the spatial regions associated with its predictions. Quantitative attention analysis was performed on the complete test set of 3,166 images.

The mean lung attention fraction was 20.7%, whereas the segmented lungs occupied approximately 24.0% of the image area. The resulting mean lung-attention enrichment was 0.868. An enrichment value of 1 indicates that attention is approximately proportional to lung area, whereas values below 1 indicate relative under-attention to the lung fields.

Class-specific results are reported in Table ??.

Table 5.3: Class-wise Grad-CAM lung attention for the full-image baseline.

Class	Lung Attention	Lung Area	Enrichment
COVID	24.46%	24.38%	1.063
Lung Opacity	11.63%	21.80%	0.518
Normal	25.24%	25.18%	1.024
Viral Pneumonia	16.81%	23.22%	0.731

The results indicate strongly class-dependent attention behaviour. COVID and Normal cases exhibited approximately proportional lung attention, whereas Lung Opacity and Viral Pneumonia showed substantially lower lung-attention enrichment.

Qualitative inspection further revealed heterogeneous attention patterns. Some predictions were associated with activations within the pulmonary fields, while others showed strong responses near image borders or outside the segmented lung regions. These findings suggest that the full-image CNN may use a combination of pulmonary information and non-pulmonary or dataset-specific visual cues.

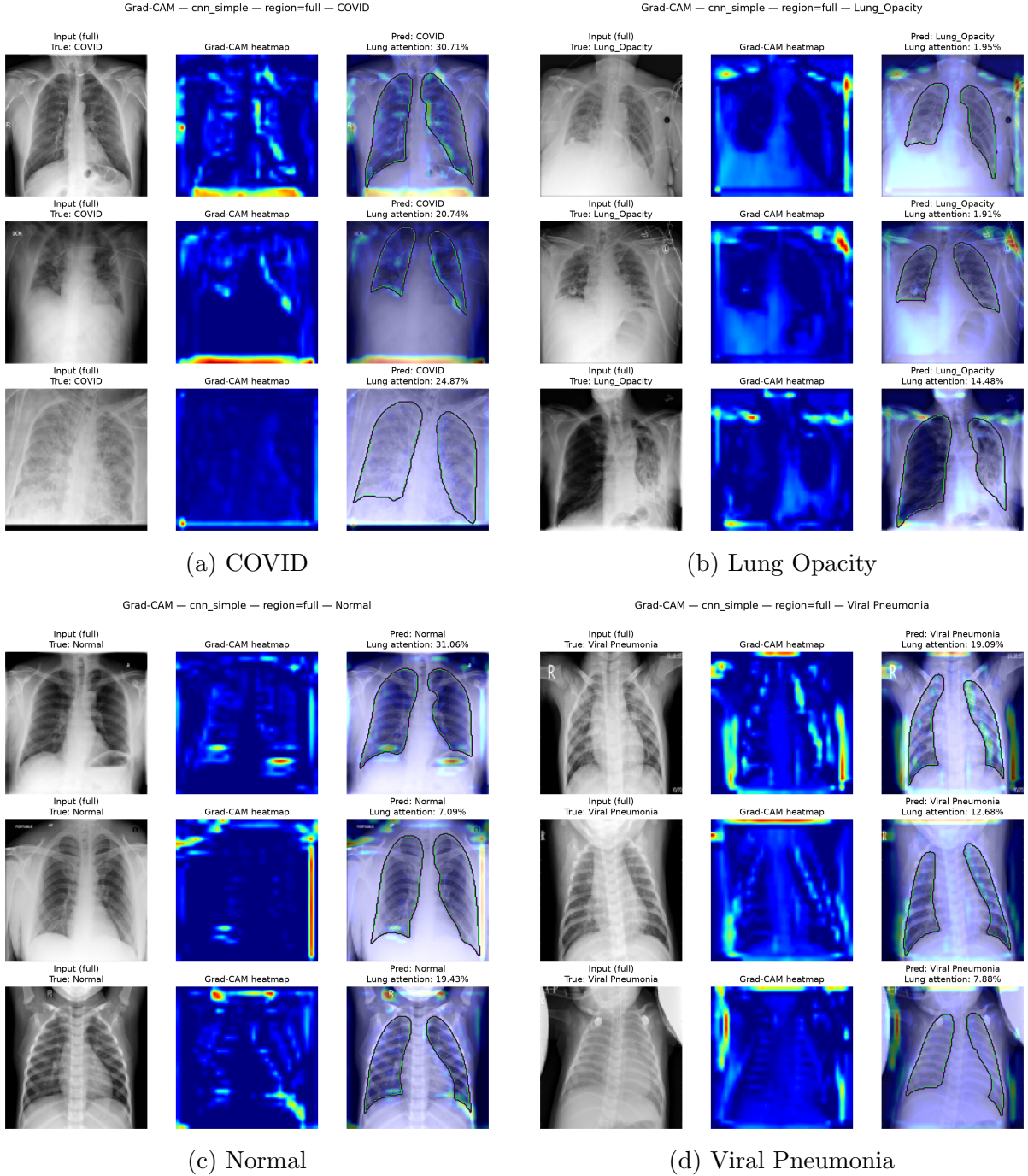


Figure 5.1: Representative Grad-CAM examples for the full lung baseline, grouped by class.

5.1.4 Motivation for a Lung-Centred ROI Baseline

A hard lung-masking experiment was initially used to restrict the CNN to the segmented pulmonary regions. This model achieved 72.93% test accuracy and 69.0% macro F1. However, hard masking introduced a sharp artificial boundary between the retained lung pixels and the zero-valued background.

Grad-CAM visualisations frequently showed activation around these segmentation boundaries. Consequently, the reduction in performance could not be attributed solely

to the removal of background information, because the masking operation itself changed the image representation and potentially introduced new geometric cues.

To obtain a cleaner controlled baseline, a lung-centred ROI preprocessing strategy was therefore introduced.

5.1.5 Baseline 2: Lung-ROI Simple CNN

For the second baseline, lung segmentation masks were used only to determine the spatial extent of the lungs. The image itself was not multiplied by the mask.

The preprocessing pipeline was:

1. Load the original chest X-ray and corresponding lung mask.
2. Use the mask to determine the joint bounding box of the two lungs.
3. Add a small spatial margin around the lung region.
4. Construct a square ROI around the lung pair.
5. Preserve the original radiographic intensities and anatomical appearance.
6. Resize the ROI to 128×128 pixels.

This preprocessing reduced variation in overall framing and lung scale while avoiding hard segmentation boundaries. No rotation normalisation, intensity normalisation, or augmentation was applied.

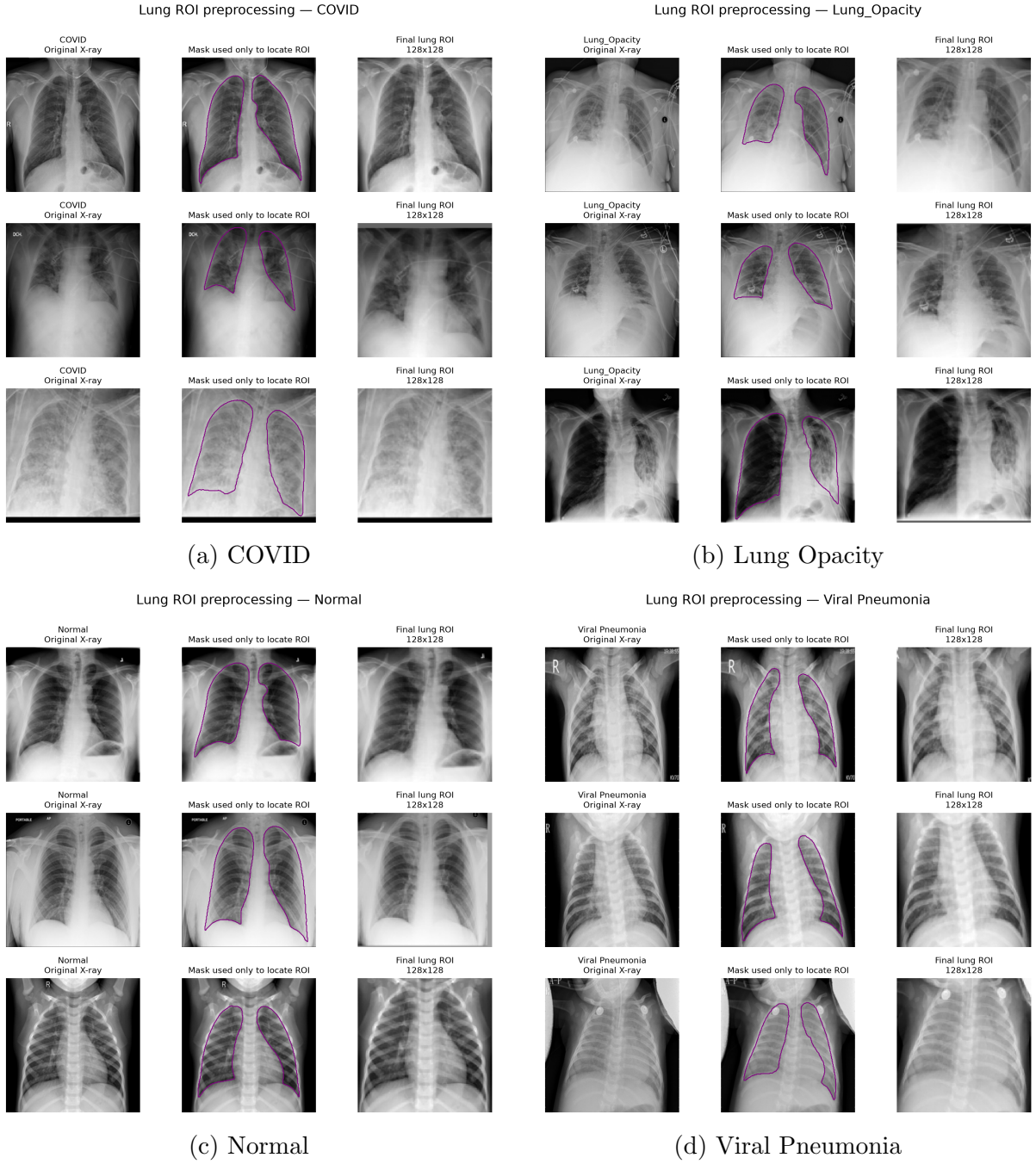


Figure 5.2: Class-wise examples of the lung-centred ROI preprocessing pipeline.

The lung-ROI model achieved the test-set performance shown in Table ??.

Table 5.4: Test-set performance of the lung-ROI simple CNN baseline.

Class	Precision	Recall	F1-score
COVID	0.649	0.703	0.675
Lung Opacity	0.822	0.691	0.751
Normal	0.828	0.853	0.840
Viral Pneumonia	0.746	0.940	0.832
Macro average	0.761	0.797	0.774

The lung-ROI model achieved **78.71% test accuracy** and **77.4% macro F1**. Its train, validation, and test accuracies were 79.86%, 78.74%, and 78.71%, respectively, indicating close agreement between the data partitions and little evidence of severe over-fitting.

5.1.6 Comparison of the Two Baselines

Table ?? compares the two principal baselines.

Table 5.5: Comparison of the two principal simple-CNN baselines.

Baseline	Test Accuracy	Macro F1
Full-image simple CNN	81.96%	82.7%
Lung-ROI simple CNN	78.71%	77.4%

Restricting the model to the lung-centred ROI reduced test accuracy by 3.25 percentage points and macro F1 by 5.3 percentage points. Nevertheless, the ROI model retained a substantial proportion of the full-image classification performance.

The class-wise recall comparison is shown in Table ??.

Table 5.6: Class-wise recall of the full-image and lung-ROI baselines.

Class	Full Image	Lung ROI
COVID	83.0%	70.3%
Lung Opacity	76.6%	69.1%
Normal	83.1%	85.3%
Viral Pneumonia	94.5%	94.0%

The largest reduction occurred for COVID classification. In contrast, Normal and Viral Pneumonia recall remained relatively stable.

These results suggest that peripheral image regions contribute to the performance of the full-image model, but they are not the sole source of discriminative information. A substantial level of classification performance remained when the input was restricted to a lung-centred thoracic ROI.

5.1.7 Grad-CAM Analysis of the Lung-ROI Baseline

Grad-CAM was also applied to the lung-ROI baseline. Unlike the hard-masked experiment, no artificial segmentation contour was present in the CNN input. Therefore, activation outside the segmented lung contour corresponds to actual anatomical or image regions contained within the ROI.

Qualitative inspection showed heterogeneous attention across the four classes. COVID examples generally demonstrated a greater degree of pulmonary activation, although attention was not exclusively confined to the lung fields. In several Lung Opacity and Viral Pneumonia examples, correctly classified images exhibited strong activation in peripheral, inferior, mediastinal, or ROI-border regions despite relatively low activation within the segmented pulmonary interior. Normal cases showed mixed behaviour, with both pulmonary and peripheral activation patterns.

These observations suggest that ROI cropping substantially reduces peripheral image information but does not completely prevent the CNN from exploiting extra-pulmonary or framing-related cues within the cropped thoracic region.

Because this ROI Grad-CAM assessment is currently based on representative qualitative examples rather than complete test-set quantitative attention analysis, the findings are interpreted as exploratory.

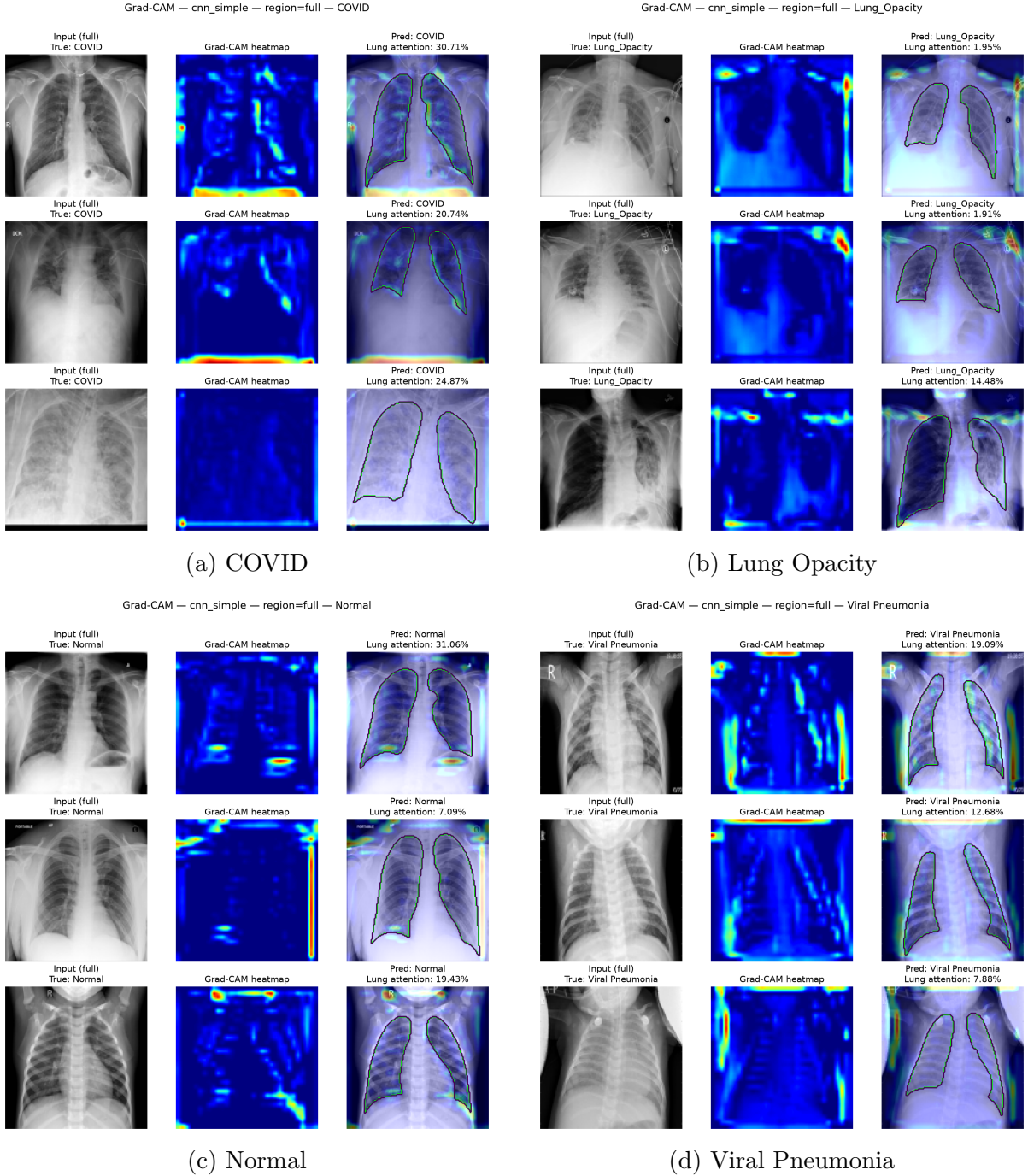


Figure 5.3: Representative Grad-CAM visualisations for the lung-ROI baseline, grouped by class. Although the ROI reduces peripheral image content and avoids hard segmentation boundaries, model attention remains heterogeneous and is not restricted exclusively to the pulmonary parenchyma.

5.1.8 Baseline Interpretation and Role in Subsequent Experiments

The two baselines provide complementary reference points for subsequent model development.

The full-image model achieved the highest baseline performance, with 81.96% test ac-

curacy and 82.7% macro F1. However, the background-only experiment and Grad-CAM analysis indicate that non-lung regions contain substantial class-correlated information. Consequently, the full-image model is retained as an unconstrained or naive performance baseline.

The lung-ROI model achieved 78.71% test accuracy and 77.4% macro F1 while substantially reducing peripheral image content and avoiding the hard-mask boundary artefact. It therefore serves as a more spatially controlled second baseline.

These configurations will be retained when evaluating subsequent augmentation and transfer-learning approaches.

This design supports two complementary questions:

1. Does a more advanced model improve overall classification performance on the complete chest radiograph?
2. Does the model also improve performance when access to obvious peripheral and background cues is reduced?

The second comparison is particularly important because an improvement in full-image accuracy alone does not necessarily imply improved use of clinically relevant pulmonary information.

Chapter 6

Results

Chapter 7

Discussion

Chapter 8

Conclusion